

Comparative Analysis of Sampling Methods in CMA-ES

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Abstract

This report presents a comparative analysis of three sampling schemes within the CMA-ES (Covariance Matrix Adaptation Evolution Strategy) algorithm: classical Gaussian sampling, Sobol sequences, and Latin Hypercube sampling. The methods are evaluated on a diverse set of standard optimization benchmarks, analyzing performance in terms of convergence, accuracy, and runtime. The results show substantial differences between methods, with practical implications for choosing sampling strategies in real-world optimization.

1 Practical notes

- Running the accompanying code typically takes about two minutes.
- GitHub repository (contributions welcome):
<https://github.com/fsampalo/Experimentacion-CMA-ES>

2 Introduction

CMA-ES is an evolutionary optimization algorithm that has proved particularly effective for continuous black-box problems. A key design choice is how candidate solutions are sampled. This study compares three sampling approaches, each with distinct properties and potential advantages.

3 Sampling methods

3.1 Classical Gaussian sampling

Classical Gaussian sampling is the traditional choice in CMA-ES. Candidates are drawn from a multivariate normal distribution whose covariance matrix is adapted during the search. Its main features are:

- Perturbations follow a normal distribution
- Covariance adaptation driven by selection
- Exploration via random perturbations in search space

3.2 Sobol sequence

The Sobol sequence is a low-discrepancy quasi-random construction that tends to cover the domain more evenly than plain random sampling. Notable properties include:

- Deterministic point generation (for a fixed counter state)
- More uniform spatial spread than i.i.d. random sampling
- Often lower variance in Monte Carlo-style estimates

3.3 Latin Hypercube

Latin Hypercube sampling stratifies each coordinate axis and draws one sample per stratum, improving marginal coverage. Benefits include:

- Better coverage of the search space
- Reduced correlation across dimensions compared with naive random sampling
- Efficient exploration in many practical settings

4 Experimental setup

4.1 Parameters

- Problem dimension: 10 (8 for the Powell function, which requires a multiple of four)
- Population size: 50
- Maximum iterations: 100
- Independent runs: 30 per method–function pair
- Search domain: $[-5, 5]$ in every coordinate

4.2 Benchmark functions

The following test functions were used:

- Sphere (simple convex bowl)
- Rosenbrock (curved valley)
- Rastrigin (many local minima)
- Ackley (high-frequency ripples)
- Schwefel (misleading landscape)
- Griewank (many suboptimal peaks)
- Levy (multimodal)
- Zakharov (quadratic with coupling)
- Dixon–Price (sensitive to scaling)
- Powell (ill-conditioned)
- Sum-Squares (weighted quadratic)
- Trid (quadratic interactions)

5 Results and analysis

5.1 Convergence behaviour

Overall, the sampling schemes differ markedly in how quickly they improve the best objective value:

- **Gaussian sampling** is more erratic and can escape shallow local structure, but in our experiments it most often achieved the weakest final values and required more iterations to reach comparable accuracy.

- **Sobol sampling** yielded the most stable and predictable improvement and, on average, the tightest final objectives. This supports the idea that low-discrepancy normal variates can be effective inside CMA-ES-style updates.
- **Latin Hypercube** offers a good compromise between exploration and exploitation: final qualities were often close to Sobol, though convergence sometimes needed more iterations.

Below we highlight a few representative runs; full comparison plots are under `imgs/comparisons`:

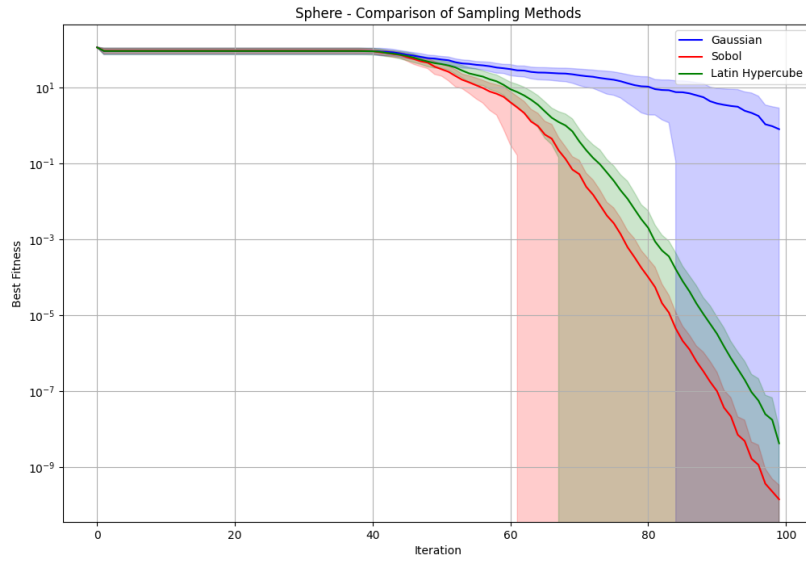


Figure 1: Convergence comparison on the Sphere function

Figure 1 shows Sobol and Latin Hypercube reaching similar best values, whereas Gaussian sampling does not reach equally accurate solutions within the same iteration budget.

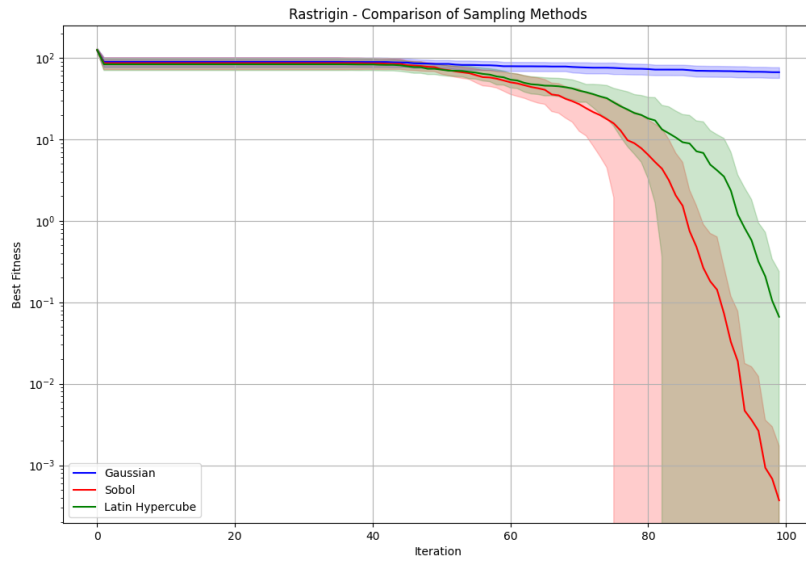


Figure 2: Convergence comparison on the Rastrigin function

Figure 2 again shows Sobol and Latin Hypercube performing comparably, in contrast to Gaussian sampling, which improves more slowly.

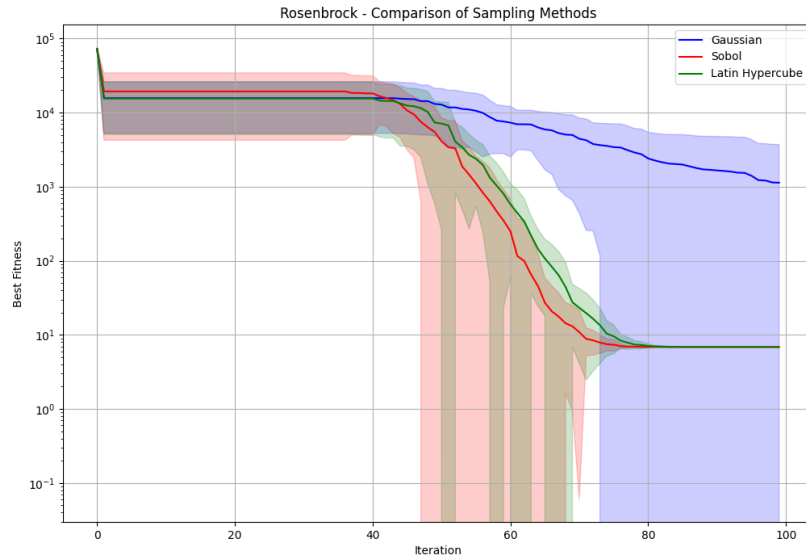


Figure 3: Convergence comparison on the Rosenbrock function

As in Figure 3, Sobol and Latin Hypercube behave similarly, while Gaussian sampling is less sample-efficient in early iterations.

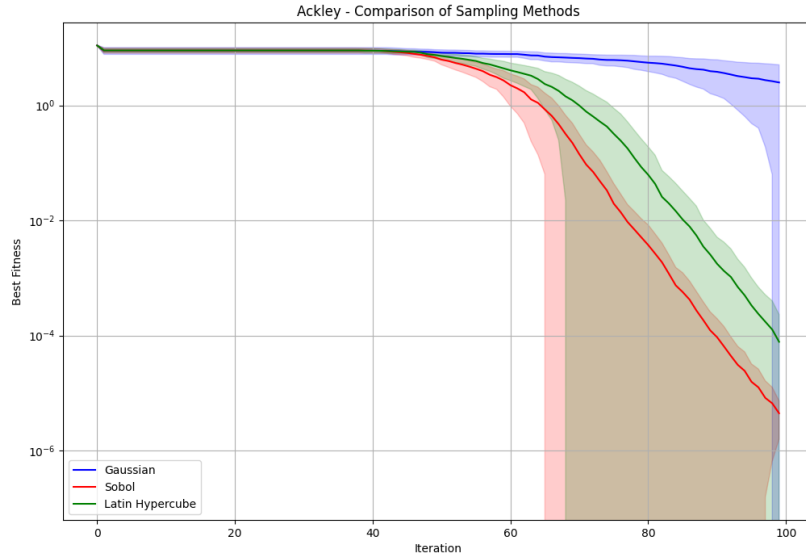


Figure 4: Convergence comparison on the Ackley function

Figure 4 highlights comparable performance for Sobol and Latin Hypercube, with Gaussian sampling initially lagging in accuracy.

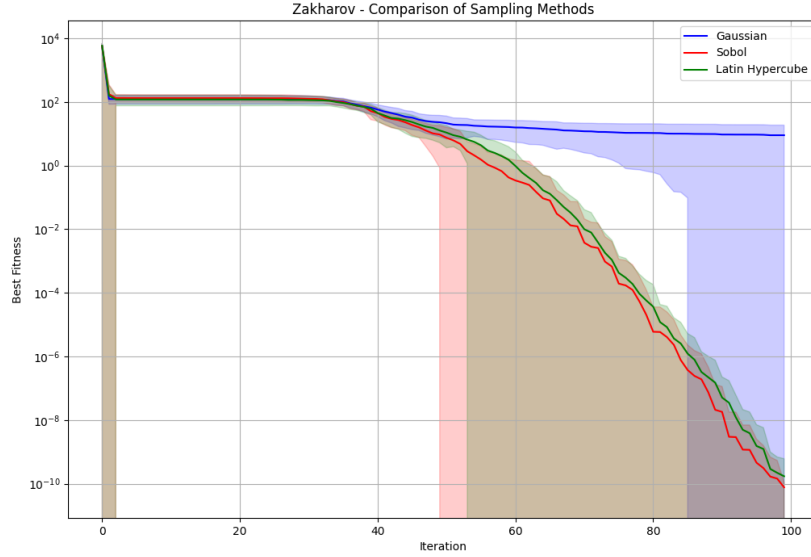


Figure 5: Convergence comparison on the Zakharov function

Figure 5 shows Sobol and Latin Hypercube converging at a similar rate, whereas Gaussian sampling needs more iterations for comparable precision.

5.2 Statistical testing

Pairwise Mann–Whitney U tests were applied to final fitness values across runs:

Function	Gaussian vs Sobol	Gaussian vs LHC	Sobol vs LHC
Sphere	0.0001	0.0001	0.7845
Rosenbrock	0.0001	0.0001	0.4119
Rastrigin	0.0001	0.0001	0.0001
Ackley	0.0001	0.0001	0.0001
Schwefel	0.3545	0.0001	0.0001

Table 1: p -values from Mann–Whitney U tests (subset of benchmarks)

We find strong evidence ($p < 0.0001$) of differences between Gaussian vs. Sobol and Gaussian vs. LHC for the listed functions, and between Sobol vs. LHC for Rastrigin, Ackley, and Schwefel. Differences between Sobol and LHC are not significant for Sphere ($p = 0.7845$) or Rosenbrock ($p = 0.4119$).

5.3 Runtime

- Gaussian sampling is usually the fastest per iteration
- Sobol sequences incur higher overhead
- Latin Hypercube sits in between

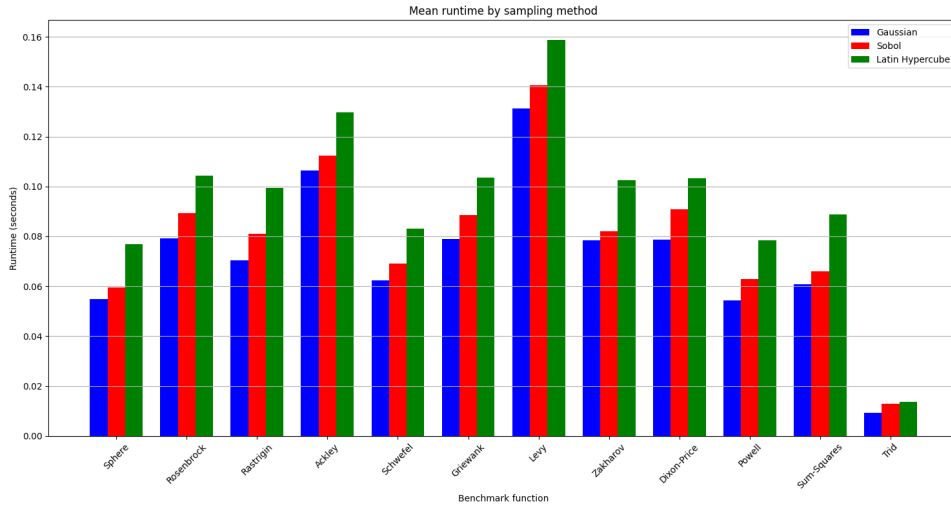


Figure 6: Runtime comparison across sampling methods

6 Discussion

Method choice should reflect:

- Problem structure (multimodality, conditioning, noise)
- Computational budget
- Whether wide exploration or fast local refinement is more important

If wall-clock speed is secondary to solution quality, Sobol and Latin Hypercube sampling are preferable in our benchmark suite.

7 Conclusions

- No single sampling rule is universally best; in our runs Sobol most often reached high-quality minima in fewer iterations.
- Sobol and Latin Hypercube excel on harder landscapes where coverage matters.
- Gaussian sampling remains simple and fast per step, but may require many more iterations on difficult functions.

8 Future work

Possible extensions include:

- Hybrid sampling schedules
- Adapting the sampler during the run
- Scaling studies in higher dimensions
- Case studies on real engineering or ML hyperparameter tasks
- Additional diversification mechanisms for highly multimodal objectives